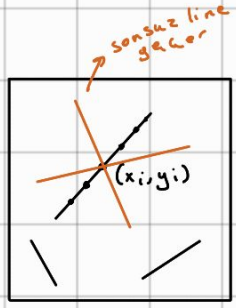
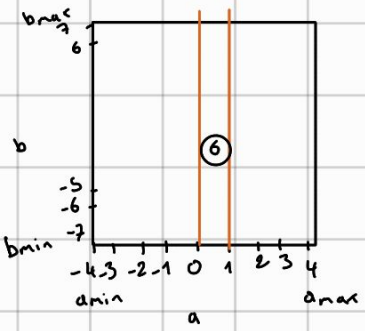
Hough Transform

Input: binary image (0,1) [edge map]



$y_i = a x_i + b$ b 0 olursa y mekezden geçer

$b = -a x_i + y_i$



doğru a yı bildiğinde aynı b'de yoğunlaşma olacak

$x_i = 1, \; y_i = 1$

$b = -a x_i + y_i$

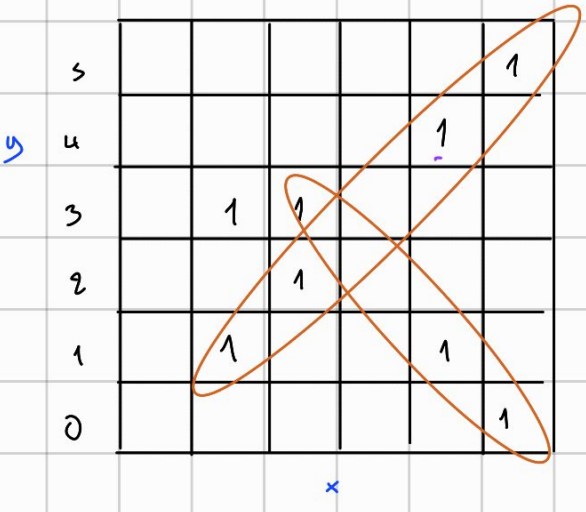
$b = -a + 1$

$a = -2 \rightarrow b = 3$

$a = -1 \rightarrow b = 2$

a	b
-2	3
-1	2
0	1
1	0
2	-1
3	-2

Örnek:



2 tane line var

$x_i = 2 \quad y_i = 3$

$b = -a x_i + y_i$

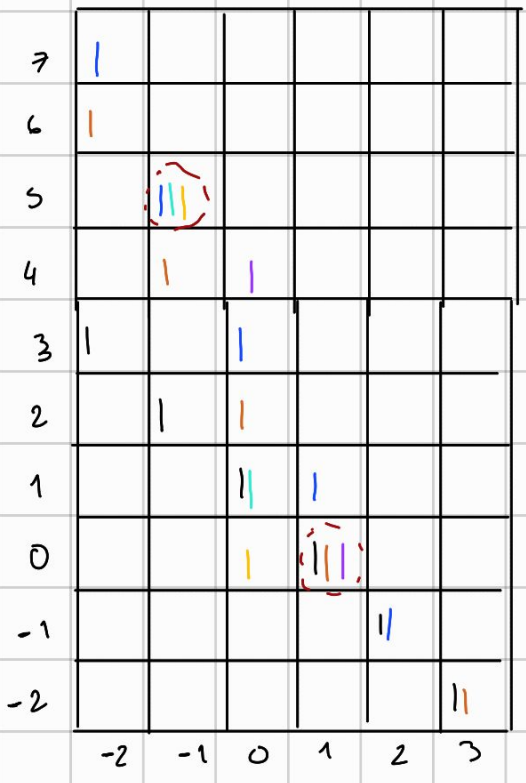
$b = -2a + 3$

$x_i = 4 \quad y_i = 1$

$b = -4a + 1$

$x_i = 5 \quad y_i = 0$

$b = -5a$



$x_i = 2, \; y_i = 2$

$b = -a x_i + y_i$

$b = -2a + 2$

a	b
-2	6
-1	4
0	2
1	0
2	-2
3	-4

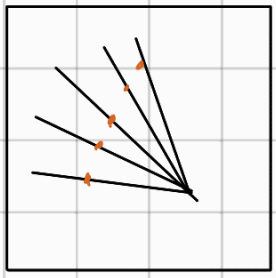
$x_i = 4 \quad y_i = 4$

$b = -a x_i + y_i$

$b = -4a + 4$

a	b
-2	12
-1	8
0	4
1	0
2	-4

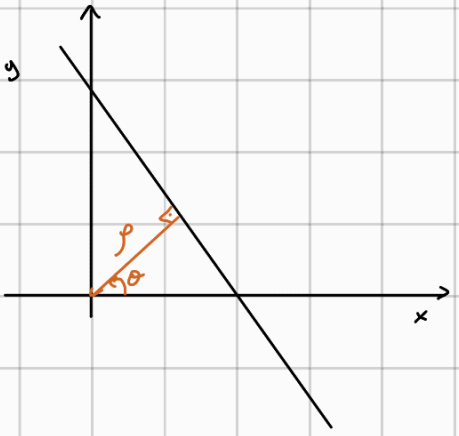
eğiği 3 yaparsak 3'ün üstü line olur



$a \to -\infty, \infty$ sıkıntı

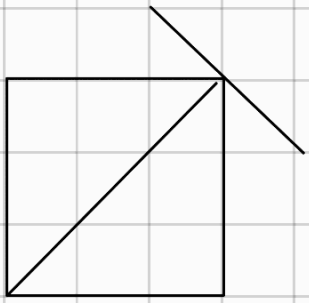
peki hangi sıklıkla

alineari olsa da ona göre açı lineer değil



$\rho = x\cos\theta + y\sin\theta$

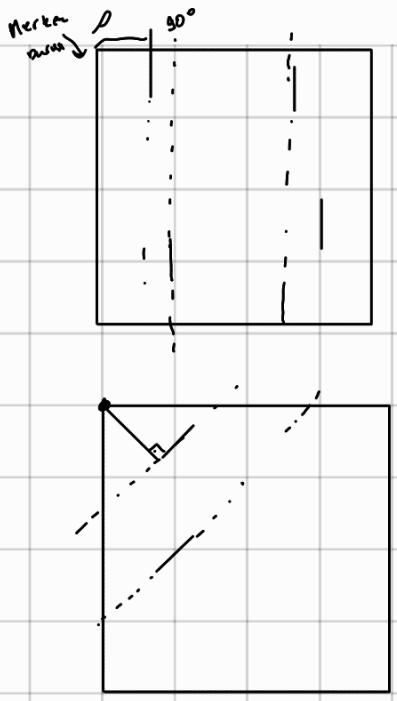
$y = ax + b$

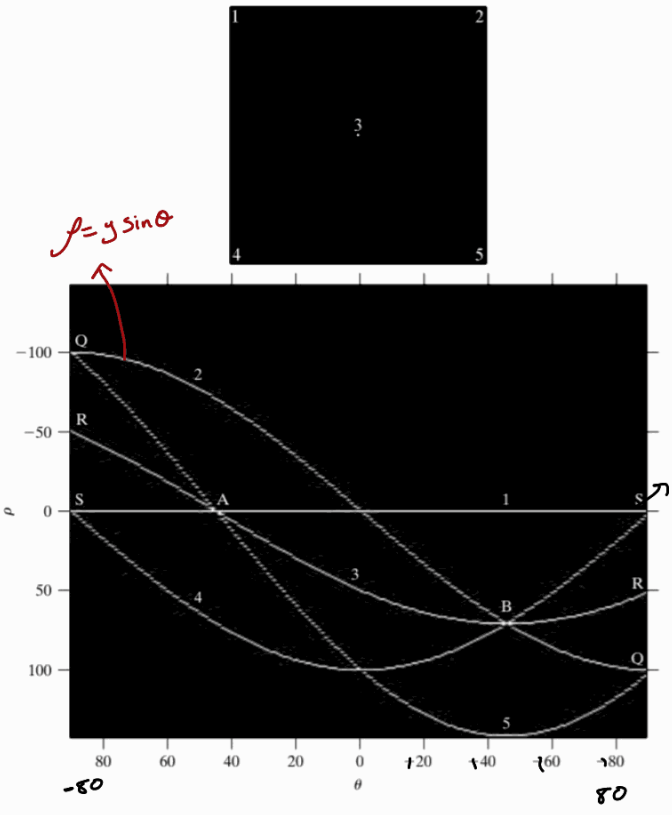


ρ ve θ uzayı daha iyi

θ $-90 + 90$ aralığında

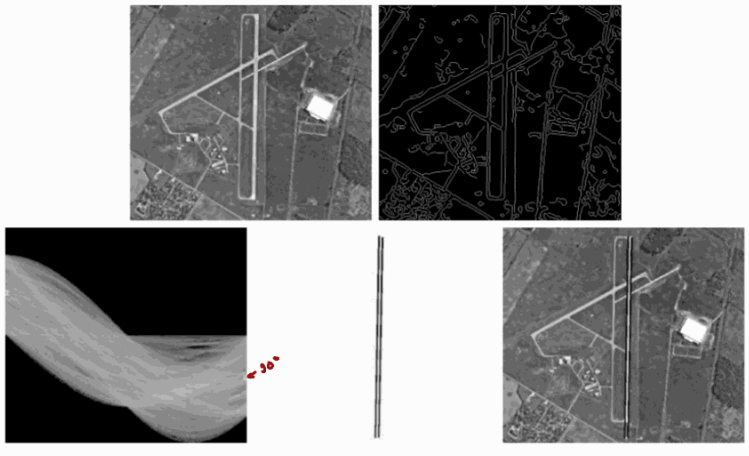
Fig 10.32





a
b

FIGURE 10.33
(a) Image of size 101 × 101 pixels, containing five points.
(b) Corresponding parameter space. (The points in (a) were enlarged to make them easier to see.)



a b
c d e

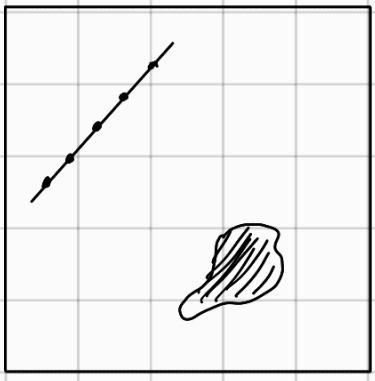
FIGURE 10.34 (a) A 502 × 564 aerial image of an airport. (b) Edge image obtained using Canny's algorithm. (c) Hough parameter space (the boxes highlight the points associated with long vertical lines). (d) Lines in the image plane corresponding to the points highlighted by the boxes). (e) Lines superimposed on the original image.

Sınavda cikar

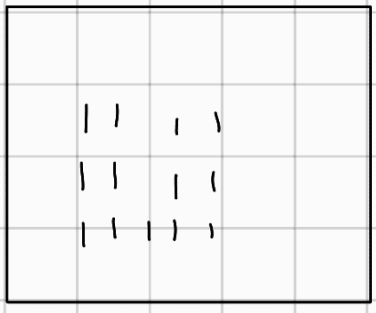
Daire için

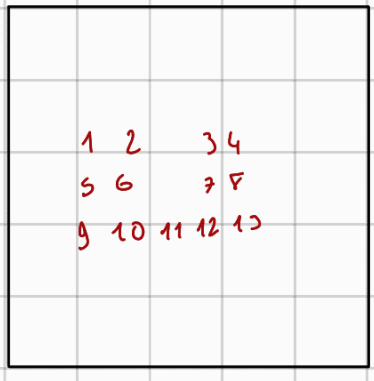
$(x-c_1)^2 + (y-c_2)^2 = c_3^2$

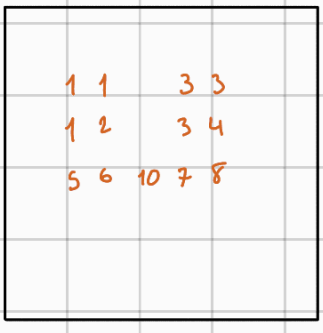
Connected Component



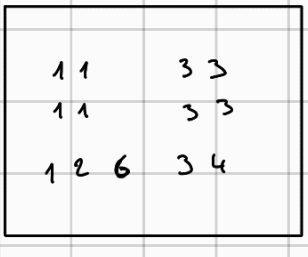
Haralick

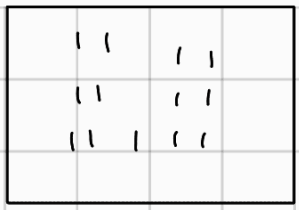












Recursive Connected Component labeling

while end of image is reached

do{ continue to scan image until
a nunlabeled "1" pixel is reached.

if(not end of image){

assign a new label to the pixel

recursively assign same label to

"1" neighbors }

1	2	3
8		4
7	6	5

soldan sağa
yukardan aşağıya

image



label

